

Requirement Analysis Document (RAD)

Project Title: E-Commerce Retail Database System (Implementation: "ShopSphere")

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1. Executive Project Summary

ShopSphere is a comprehensive online retail platform that allows users to browse product catalogs, manage shopping carts, place orders, process secure payments, and track shipments. Managing e-commerce transactions in real-time requires a robust database network: when a customer checks out, the specific inventory items must be locked immediately to prevent overselling, the payment gateway must be verified, and order statuses must be systematically updated.

To manage this safely, ShopSphere relies on a Relational Database Management System (RDBMS) like PostgreSQL or Oracle. A relational database is critical because it enforces strict data rules and guarantees (Atomicity, Consistency, Isolation, Durability) transaction properties. For example:

- **Atomicity (All-or-Nothing):** If a user checks out, the database must both record the payment and deduct the purchased items from inventory. If the payment fails halfway through, the entire transaction is rolled back so the inventory remains untouched, and the user is not charged.
- **Consistency (Valid State):** The system prevents invalid states. A product price cannot accidentally be saved as negative (< 0), and an order cannot be placed for an item that is completely out of stock.

2. Core Business Domains & Features

Based on our analysis of online retail operations, we have divided the database requirements into four distinct functional modules:

2.1 User & Account Management

This module handles customer profiles, login credentials, shipping addresses, and role management.

- **Business Rules:**
 - Every user must provide a unique email address during registration. This email acts as their login credential and cannot be shared.

- A user is assigned a unique UserID upon registration, alongside their Name, Contact Number, and a system-generated Registration Date.
- Users can save multiple Shipping Addresses to their profile, tagging one as the "Default" address.
- A user can exist in the system without having placed any orders yet (e.g., browsing or cart creation).

2.2 Product Catalog & Inventory Management

This module houses the database records for product categories, item details, pricing, and warehouse stock levels.

- **Business Rules:**

- Products are organized into "Categories" (e.g., Electronics, Apparel). Every product must belong to at least one primary Category.
- Each Product has a unique SKU (Stock Keeping Unit), a descriptive Name, a Price, and an associated Stock Quantity.
- The database must enforce the constraint that Stock Quantity cannot drop below zero (≥ 0).
- If a product's stock reaches zero, its availability status must automatically toggle to "Out of Stock".

2.3 Shopping Cart & Order Lifecycle

This module acts as the core transaction engine, capturing cart sessions, order details, and fulfillment progress.

- **Business Rules:**

- A single order is identified by a unique OrderID. It is linked to exactly one Customer.
- An order can contain multiple different products. This creates a many-to-many (M:N) relationship between Order and Product, which is resolved using a bridging table called OrderItems.
- **The Historic Price Rule:** The unit price recorded in OrderItems must represent the price at the exact second of purchase. If the store decides to run a 50% sale next week, past purchase receipts must remain unchanged to preserve correct accounting.
- Orders must support a rigid lifecycle state machine: Pending -> Payment Confirmed -> Shipped -> Delivered (or Cancelled).

2.4 Payments & Shipping Operations

This module manages currency validation, shipping carrier integration, and tracking logistics.

- **Business Rules:**

- Each Order has a single Payment profile. A Payment cannot progress without a corresponding valid OrderID (1:1 relationship).
- An order status cannot progress to Shipped until its corresponding Payment status is recorded as Successful.
- Once shipped, the system must generate a unique Tracking Number linked to the Order. The order status changes to Delivered only when the shipping carrier's API confirms delivery.

3. Actor Profiles & System Interactions

To maintain system security, different users must have distinct database access privileges. We define four primary actors:

1. Customer (Shopper)

- *What they can Read:* Product catalogs, prices, their own order history, and shipping statuses.
- *What they can Write/Edit:* Their profile info, shipping addresses, shopping cart contents, and product reviews.
- *Safety Boundary:* A Customer can never view another user's personal details, payment methods, or order history. They cannot alter product prices or manipulate stock levels.

2. Store Manager

- *What they can Read:* Sales statistics, inventory levels, and category performance.
- *What they can Write/Edit:* Add new products to the catalog, update prices, adjust stock quantities, and mark items as "Discontinued".
- *Safety Boundary:* Managers cannot view customer credit card data, manually override transactional payment gateways, or delete historical order audit logs.

3. Fulfillment Staff (Logistics)

- *What they can Read:* Order details, shipping addresses, and product packing instructions.
- *What they can Write/Edit:* Update Order Status from Payment Confirmed to Shipped, and input Tracking Numbers.
- *Safety Boundary:* They cannot view customer financial records, edit product prices, cancel orders, or alter catalog data.

4. System Admin

- *What they can Read/Write:* Complete (Create, Read, Update, Delete) privileges across all database tables. All Admin transactions are logged to an audit table to prevent internal fraud.

4. Analytical Reporting Requirements

To help ShopSphere's management optimize retail performance, the database must be structurally capable of executing aggregate joins and calculations to produce these three business reports:

- **Report 1: Low-Inventory Restock Alerts**

- *Business Value:* Identifies products that need immediate restocking to prevent lost sales.
- *Logical Query Goal:* Find all active products where the current stock quantity is less than or equal to a critical threshold (k , where $k = 10$). The report must list the Product Name, SKU, Category, Current Stock, and Supplier Contact Info.

- **Report 2: VIP Customer Loyalty Report**

- *Business Value:* Allows the marketing team to distribute discount coupons to high-spending users.
- *Logical Query Goal:* Retrieve the top N (where $N = 50$) of all registered customers ranked by their total successful order dollar volume over the past 12 months. The report must list the Customer's Name, Email Address, and total amount spent.

- **Report 3: Category Sales Performance Directory**

- *Business Value:* Shows which product categories generate the most profit, guiding future inventory purchasing.
- *Logical Query Goal:* For a specified calendar month, calculate the total revenue generated, the total number of items sold, and the average product review rating for each distinct Category.

5. Non-Functional Requirements & Database Constraints

Our system rules ensure the database is secure, performant, and reliable under real-world conditions:

- **Strict Data Constraints:**

- Product review ratings must be integers between 1 and 5 inclusive (CHECK (Rating ≥ 1 AND Rating ≤ 5)).
- Product prices and order amounts must be strictly positive values (> 0).
- The checkout system must enforce that a customer cannot purchase a quantity greater than the available stock.

- **Concurrency Controls:**

- If two users click "Buy" for the last available item at the exact same millisecond, the database must use Row-Level Transaction Locks (SELECT FOR UPDATE). The first user's

request will lock the specific inventory row, process their payment, deduct the stock, and release the lock. The second user's transaction will then read the updated state (Stock = 0) and gracefully inform them that the item is out of stock.

- **Historical Auditing & Safe Deletion:**

- To comply with financial accounting and data privacy regulations, user accounts cannot be permanently hard-deleted. If a user deletes their account, their profile status is updated to Inactive, and personal identifiers (like saved credit cards and phone numbers) are blanked out for privacy compliance. However, their past order numbers and receipts must remain in our database to ensure that financial audits and historical sales reports balance perfectly.